

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 97-033

WASTE DISCHARGE REQUIREMENTS
FOR
U.S. DEPARTMENT OF INTERIOR, NATIONAL PARK SERVICE
SEQUOIA-KINGS CANYON NATIONAL PARK
CEDAR GROVE WASTEWATER TREATMENT FACILITY
FRESNO COUNTY

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Board) finds that:

1. U.S. Department of Interior, National Park Service, Sequoia-Kings Canyon National Park, (hereafter Discharger) discharges domestic waste from a wastewater treatment facility in the Cedar Grove area of the park.
2. The wastewater treatment facility operates for five to six months each year to provide treatment for wastes from campgrounds, employee housing, and concessions during the May through November use season. Currently, the monthly average flow at the facility is about 19,500 gallons per day (gpd). The daily maximum flow is about 55,000 gpd. The wastewater treatment facility is designed for a daily average flow of 55,000 gpd.
3. Waste Discharge Requirements Order No. 80-154, adopted by the Board on 24 October 1980, prescribes requirements for the facility. The facility currently has an extended aeration package plant for treatment, and two evaporation/percolation ponds for disposal.
4. The Discharger submitted a Report of Waste Discharge on 13 April 1995 to reflect planned changes in the method of treatment and disposal at the facility. The Discharger has problems with solids at the facility because of insufficient control mechanisms. Also, the disposal ponds are very close to the Kings River with potential threat of impacts to the river.
5. The Discharger proposes to modify the treatment and disposal system at the facility. The Discharger will continue to treat in its existing aeration tank and discharge the wastewater to the disposal ponds until construction is completed. Then the existing aeration tank will be modified into a septic tank to provide primary treatment. A recirculating slow rate sand filter will be constructed northwest of the existing facility.

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for further treatment of wastewater. Spent sand from the filters will be backwashed in place and reused. The Discharger proposes future discharge of the effluent through a leachfield system consisting of three fields. The modified wastewater treatment facility will be designed to treat and dispose an average flow of 55,000 gpd.

6. The heavy warm rains of late December 1996 damaged the road serving the Cedar Grove area. The Discharger's schedule for the modifications is to begin construction of the modifications by 1 July 1997, and this schedule is reflected in Provision E.4. The access road may not be open by this date, which would delay beginning construction.
7. Sludge from the septic tank is proposed to be hauled to municipal wastewater treatment plants in the valley through contracts with licensed hauling companies.
8. Order No. 80-154 does not reflect the modified wastewater treatment and disposal methods and is not consistent with current plans and policies of the Board.
9. Soils in the area consist of silty sands extending greater than 9.5 feet. The soils contain occasional cobbles up to 12 inches and boulders up to 18 inches in diameter. Average ground slope is 12%.
10. The facility is in Sections 11 and 14, T13S, R30E, MDB&M, as shown in Attachment A, which is part of this Order by reference. Surface water drainage is to the South Fork of the Kings River. The site lies within the South Fork Kings Hydrologic Subarea of the Upper Kings Hydrologic Area (No. 552.34) in the Kings River Hydrologic Unit, as depicted on interagency hydrologic maps prepared by the Department of Water Resources in August 1986.
11. The *Water Quality Control Plan for the Tulare Lake Basin, Second Edition*, (hereafter Basin Plan) designates beneficial uses, establishes water quality objectives, and contains implementation plans and policies for waters of the basin. These requirements implement the Basin Plan.
12. The beneficial uses of the Kings River above Kirch Flat, as identified in the Basin Plan, include municipal supply, water contact and non-contact water recreation, warm and cold fresh water habitat, wildlife habitat, preservation of rare and endangered species, fish spawning, and fresh water replenishment.

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13. The beneficial uses of ground water in the area are domestic, industrial, and agricultural supply.
14. The U.S. Department of Interior, National Park Service, completed an environmental assessment for the project and issued a Finding of No Significant Impact on 12 December 1996. The Board circulated the environmental assessment and a completed checklist to other agencies through the State Clearinghouse. The project is to replace or modify an existing treatment facility within a national park, and the park has a policy which does not allow growth. There will be no growth-inducing impacts. The environmental assessment by the Discharger, coupled with circulation of the assessment by the Board, satisfies the requirements of the California Environmental Quality Act (CEQA) (Public Resources Code Section 21000, et seq.) The Board has reviewed the environmental assessment and concurs there are no significant impacts on water quality.
15. The Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.
16. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Order No. 80-154 is rescinded and U.S. Department of Interior, National Park Service, Sequoia-Kings Canyon National Park, its agents, successors, and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following:

A. Discharge Prohibitions:

1. Discharge of waste to surface waters or surface water drainage courses is prohibited.
2. Bypass or overflow of untreated or partially treated waste is prohibited except as allowed in Provision E.2 of Standard Provisions and Reporting Requirements.
3. Discharge of waste classified as 'hazardous' or 'designated', as defined in Sections 2521(a) and 2522(a) of Title 23, CCR, Section 2510, et seq., (hereafter Chapter 15) is prohibited.

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B. Discharge Specifications:

1. The monthly average discharge shall not exceed 55,000 gpd.
2. Objectionable odors originating at this facility (both as it exists and after modification) shall not be perceivable beyond the limits of the wastewater treatment and disposal area.
3. As one means of discerning compliance with Discharge Specification No. B.2, the dissolved oxygen content in the upper zone (1 foot) of wastewater in ponds shall not be less than 1.0 mg/l.
4. The facility shall be designed, constructed, operated, and maintained to prevent inundation or washout due to floods with a 100-year return frequency.
5. Ponds shall not have a pH less than 6.5 or greater than 8.5.
6. Ponds shall be managed to prevent breeding of mosquitoes. In particular,
 - a. An erosion control program should assure that small coves and irregularities are not created around the perimeter of the water surface.
 - b. Weeds shall be minimized through control of water depth, harvesting, or herbicides.
 - c. Dead algae, vegetation, and debris shall not accumulate on the water surface.
7. Ponds shall have sufficient capacity to accommodate allowable wastewater flow and design seasonal precipitation and ancillary inflow and infiltration during the non-irrigation season. Design seasonal precipitation shall be based on total annual precipitation using a return period of 100 years, distributed monthly in accordance with historical rainfall patterns. Freeboard shall never be less than two feet (measured vertically).
8. On or about 1 October of each year, available pond storage capacity shall be at least equal to the volume necessary to comply with Discharge Specification B.7.

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9. The maximum conductivity (specific electrical conductance at 25° C, also 'EC') of the discharge shall not exceed the source water EC plus 300 μ mhos/cm.
10. Public contact with wastewater shall be precluded through such means as fences and signs, or acceptable alternatives.
11. The minimum horizontal distance between the leachfields and the edge of drainage courses or ephemeral streams shall be 50 feet.
12. The minimum horizontal distance between the leachfields and the 10-year frequency flood line of flowing streams shall be 100 feet.

C. Sludge Disposal Specifications:

1. Collected screenings, sludges, and other solids removed from liquid wastes shall be disposed of in a manner that is consistent with Chapter 15 and approved by the Executive Officer.
2. Any proposed change to initiate sludge use or disposal from the current transport to a municipal wastewater treatment plant shall be reported to the Executive Officer at least 90 days in advance of the change.

D. Ground Water Limitations:

The discharge, in combination with other sources, shall not cause underlying ground water to contain waste constituents in concentrations statistically greater than background water quality.

E. Provisions:

1. The Discharger shall comply with Monitoring and Reporting Program No. 97-033, which is part of this Order, and any revisions thereto as ordered by the Executive Officer.

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2. The Discharger shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements," dated 1 March 1991, which are attached hereto and by reference a part of this Order. This attachment and its individual paragraphs are commonly referenced as "Standard Provision(s)."
3. The Discharger must comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Regional Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.
4. The Discharger shall comply with the following time schedule:

<u>Task</u>	<u>Compliance Date</u>	<u>Report Date</u>
a. Submit plans and design details of the proposed treatment and disposal units.	2 Mar 1997	2 Mar 1997
b. Begin Construction	1 Jul 1997	15 Jul 1997
c. Submit written status report	1 Dec 1997	1 Dec 1997
d. Complete construction	1 Jul 1998	15 Jul 1998
e. Remove all sludge accumulated in the disposal ponds.	1 Dec 1998	15 Dec 1998

All technical reports shall be prepared under the direction of a qualified professional engineer. All reports are subject to the review and approval of the Executive Officer.

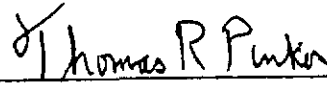
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If the WWTF is inaccessible by 1 June 1997 because of the wet winter such that construction cannot begin by 1 July 1997, then following notification from the Discharger the Executive Officer may, at his discretion to account for the delay, change the compliance dates for 4.b. & 4.d. listed above. The compliance date for 4.b. and 4.d. shall not be extended more than six months without bringing the matter to the Board's attention.

5. By 15 July 1998, the Discharger shall submit a copy of the operations and maintenance manual (O&M). The O&M shall be prepared by a qualified professional engineer and shall include monitoring and maintenance procedures for the new treatment and the disposal system.
6. The Discharger must comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Regional Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.
7. A copy of this Order shall be kept at the discharge facility for reference by wastewater treatment plant operating personnel. Key operating personnel shall be familiar with its contents.
8. The Board will review this Order periodically and will revise requirements when necessary.

I, JAMES R. BENNETT, Interim Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 28 February 1997.


for JAMES R. BENNETT, Interim Executive Officer

SH:sh/fmc:2/28/97

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

MONITORING AND REPORTING PROGRAM NO. 97-033

FOR
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Specific sample station locations shall be established with concurrence of the Board's staff, and a description of the stations shall be submitted to Board and attached to this Program.

INFLUENT MONITORING

During the operational season, influent samples shall be collected at the inlet of the headworks and approximately the same time as an effluent sample. Influent monitoring shall include the following:

<u>Constituent</u>	<u>Units</u>	<u>Measurement</u>	<u>Frequency</u>
Flow	mgd	Metered	Continuously

EFFLUENT MONITORING

During the operational season, effluent samples shall be collected at the outlet of the wastewater treatment facility or the recirculating sand filter prior to discharge to the ponds or leachfields. Effluent samples shall be representative of the volume and nature of the discharge. Time of collection of a grab sample shall be recorded. Effluent monitoring shall include the following:

<u>Constituent</u>	<u>Units</u>	<u>Measurement</u>	<u>Frequency</u>
EC	μ mhos/cm	Grab	Monthly
BOD ¹	mg/l	Grab	Monthly ²

1 5-day, 20 °C biochemical oxygen demand.

2 BOD of the effluent from the wastewater treatment facility shall be monitored until the ponds are no longer in use.

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SEPTIC TANK MONITORING

The septic tank shall be inspected at least annually, and pumped when appropriate. The dates and nature of inspections and maintenance shall be included in the monitoring reports. In addition, effluent from the septic tanks, prior to treatment in the filter, shall be monitored every month for total suspended solids and the results included in the monitoring reports.

POND MONITORING

Pond monitoring shall continue seasonally until the ponds are no longer in use. The freeboard shall be monitored on the evaporation/percolation ponds in use to the nearest foot. Pond water monitoring shall include the following:

<u>Constituent</u>	<u>Unit</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
Freeboard	feet	Estimate	Weekly
Dissolved Oxygen ¹	mg/l	Grab	Weekly

¹ Samples shall be collected at a depth of one foot from each pond, opposite the inlet, and analyzed for dissolved oxygen. Samples shall be collected between 0800 and 0900 hours.

In addition, the Discharger shall inspect the condition of the ponds once per week and write visual observation in a bound log book. Notations shall include observations of whether weeds are developing in the water or along the bank, and their location; whether dead algae, vegetation, scum, or debris are accumulating on the pond surface and their location; whether burrowing animals or insects are present; and the color of the pond (e.g., dark sparkling green, dull green, yellow, grey, tan, brown, etc.). A copy of the entries made in the log during each month shall be submitted along with the monitoring report the following month. Where the O&M manual indicates remedial action is necessary, the Discharger shall briefly explain in the transmittal what action has been taken or is scheduled to be taken.

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LEACHFIELD MONITORING

During the operational season, the Discharger shall monitor and visually inspect the leachfields once per week. In conducting the review, a log shall be kept of leachfield conditions. The presence or absence of surfacing effluent or back-up of wastewater in leachlines shall be noted.

Prior to commencing discharge to the new leachfield, a minimum of four representative piezometers or monitoring wells shall be established at the leachfield to monitor the vertical proximity of ground water within the soil zone extending to five feet beneath the trench bottoms in the leachfield. Prior to installation of the piezometers or the monitoring wells, the Discharger shall submit a technical report, subject to the approval of the Executive Officer, providing (1) a drawing showing the location of the piezometers or monitoring wells; (2) construction details of the piezometer construction or the monitoring well construction; (3) the depth from the ground surface to leaching trench bottoms. Leachfield monitoring shall include the following:

<u>Constituent</u>	<u>Units</u>	<u>Measurement</u>	<u>Frequency</u>
Depth ¹	feet	Observation	Weekly ²

¹ Depth shall be measured to the nearest tenth of a foot

² Observations shall be made during the operational season

REPORTING

Until the use of ponds ceases, daily, weekly, and monthly monitoring data shall be reported in monthly monitoring reports. Monthly monitoring reports shall be submitted to the Board by the **20th day of the following month**. Following completion of construction of modified treatment facilities and leachfields and termination of use of the disposal ponds, all monitoring data shall be submitted to the Board in an annual report due by **1 December** of each year.

In reporting the monitoring data, the Discharger shall arrange the data in tabular form so that the date, the constituents, and the concentrations are readily discernible. The data shall be summarized in such a manner that illustrates clearly whether the Discharger complies with waste discharge requirements.

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If the Discharger monitors any pollutant at the locations designated herein more frequently than is required by this Order, the results of such monitoring shall be included in the discharge monitoring report.

The Discharger may also be requested to submit an annual report to the Board with tabular and graphical summaries of the monitoring data obtained during the previous year. Any such request shall be made in writing. The report shall discuss the corrective actions taken and planned to bring the discharge into full compliance with the waste discharge requirements.

By 1 December of each year, the Discharger shall submit a written report to the Executive Officer containing the following:

- a. The names, certificate grades (if any), and general responsibilities of all persons in charge of wastewater treatment and disposal (Standard Provision A.5).
- b. The names and telephone numbers of persons to contact regarding wastewater disposal for emergency and routine situations.
- c. A statement certifying when the flow meter and other monitoring instruments and devices were last calibrated, including identification of who performed the calibration (Standard Provision C.6).
- d. A statement certifying whether the current operation and maintenance manual, and contingency plan, reflect the wastewater treatment plant as currently constructed and operated, and the dates when these documents were last revised and last reviewed for adequacy.

All reports submitted in response to this Order shall comply with the signatory requirements in Standard Provision B.3.

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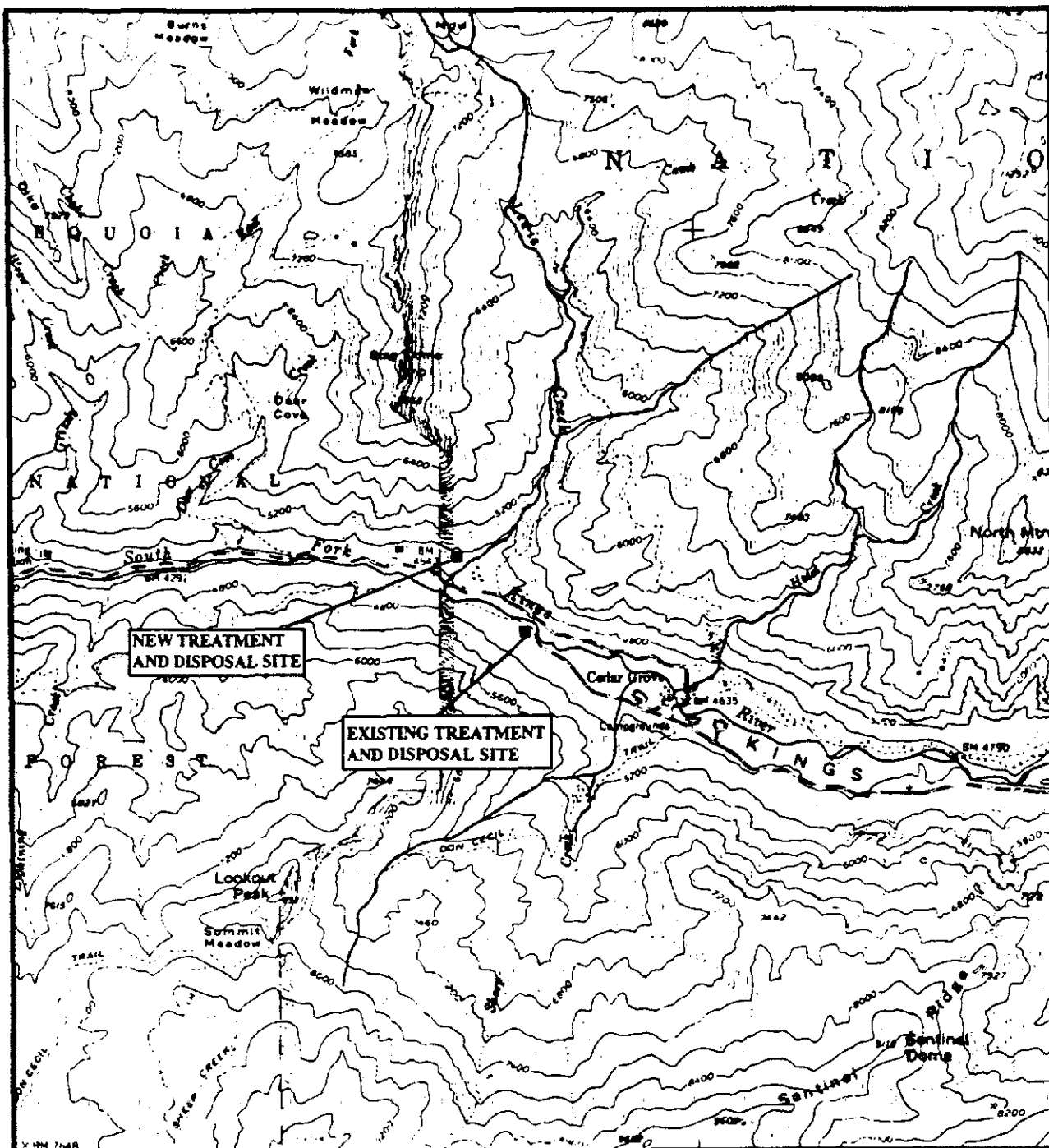
The Discharger shall implement the above monitoring program on the first day of the month following adoption of this Order.

Ordered by: Thomas R. Pinkis
for JAMES R. BENNETT, Interim Executive Officer

28 February 1997

(Date)

SH:sh/fmc:2/28/97



ATTACHMENT A

U.S. DEPARTMENT OF INTERIOR, NATIONAL PARK SERVICE
 SEQUOIA-KINGS CANYON NATIONAL PARK
 CEDAR GROVE WASTEWATER TREATMENT FACILITY
 FRESNO COUNTY

SECTIONS 11 AND 14, T13S, R30E, MDB&M
 MARION PEAK QUAD, 15 MINUTE SERIES

SCALE 1 : 62,500

INFORMATION SHEET

U. S. DEPARTMENT OF INTERIOR, NATIONAL PARK SERVICE SEQUOIA-KINGS CANYON NATIONAL PARK CEDAR GROVE WASTEWATER TREATMENT FACILITY FRESNO COUNTY

U. S. Department of Interior, National Park Service, operates a wastewater treatment and disposal facility at the Cedar Grove area in the Sequoia-Kings Canyon National Park. The facility is currently regulated by Waste Discharge Requirements Order No. 80-154. The facility is about 30 miles upstream of Pine Flat Reservoir. The facility serves four campgrounds with over 400 campsites, offices, workshops, storage facilities, employee residences, a concession store, a lunch counter, and eight overnight accommodations. Currently, the monthly average flow at the facility is about 19,000 gallons per day (gpd). The daily maximum flow is about 55,000 gpd. The facility is located at 4800-foot elevation and includes headworks with a communitor and a bar screen, an extended aeration package treatment plant and two evaporation/percolation ponds.

The package treatment plant consists of a 40-foot diameter concrete tank with a steel inner ring of 20-foot diameter. The outer portion is for aeration and the inner circle is the clarifier. The major problem with the existing plant is that there are insufficient control mechanisms for dealing with the solids. A water meter is currently installed in the effluent line to measure the flow leaving the plant. Solids currently pass through this meter, clogging it at times, causing the measured flow values to be inaccurate. The original steel components of the existing treatment system are rusted and need replacement or repair. The existing disposal ponds are very close to the South Fork of the Kings River.

The Discharger proposes to change the treatment and disposal. The existing package plant will be modified into a septic tank to provide primary treatment. A recirculating slow rate sand filter is proposed to be constructed, northwest of the existing treatment system, for further treatment of wastewater. Spent sand from the filters will be backwashed at the site and reused. The Discharger proposes to dispose the effluent through a leachfield system consisting of three fields. The Discharger will continue to treat wastewater in its existing aeration tank and discharge the wastewater to the disposal ponds until construction is completed.

Sludge from the septic tank will be hauled to municipal wastewater treatment plants in the valley through contracts with licensed hauling companies.

The discharge specifications and monitoring program are revised to reflect changes in the method of treatment and disposal at the facility. This order includes provisions with a time schedule for constructing the new treatment and disposal units.

Cedar Grove is closed during the winter. The facility usually operates from May to September, but in some years it operates until November, weather permitting.

INFORMATION SHEET - Continued

U.S. DEPARTMENT OF INTERIOR, NATIONAL PARK SERVICE
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Heavy warm rains of late December 1996 damaged the road accessing the Cedar Grove area. The Discharger is optimistic that the road will be open by 1 June 1997. However, if the road is not open by this date, the date for beginning construction of 1 July 1997 will be delayed. The order grants the Executive Officer authority to extend the construction compliance dates by up to six (6) months if the road is not open by 1 June 1997.

Surface water drainage from the facility is to the South Fork of the Kings River. The beneficial uses of the Kings River above Kirch Flat, as identified in the Basin Plan, include water contact and non-contact water recreation, warm and cold fresh water habitat, wildlife habitat, preservation of rare and endangered species, fish spawning, and fresh water replenishment.

Soils in the area consist of silty sands extending greater than 9.5 feet. The soils contain occasional cobbles up to 12 inches and boulders up to 18 inches in diameter. Average ground slope is 12%.

Precipitation in the area is about 30 inches per year.

The U.S. Department of Interior, National Park Service completed an environmental assessment for the project and issued a Finding of No Significant Impact on 12 December 1996. The Board circulated the environmental assessment and a completed checklist to other agencies through the State Clearinghouse. The project is to replace or modify an existing treatment facility within a national park, and the park has a policy which does not allow growth. There will be no growth-inducing impacts. The environmental assessment by the Discharger, coupled with circulation of the assessment by the Board, satisfies the requirements of the California Environmental Quality Act (CEQA) (Public Resources Code Section 21000, et seq.) The Board has reviewed the environmental assessment and concurs there are no significant impacts on water quality.

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